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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CV2251-1 – AIRPORTS AND SEAPORTS ENGINEERING
(For CV)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which factor is a primary contributor to the escalation in global air traffic?
A) Increased government subsidies for airlines B) The rise of high-speed rail transportation
C) Globalization and the proliferation of low-cost airlines D) The decline in aircraft manufacturing
2. Which of the following is NOT one of the key stages in the airport development process?
A) Forecasting future traffic volumes B) Rigorously assessing and improving existing airport capacity
C) Designing a detailed employee training program D) Planning for the establishment of entirely new airports
3. What does the "ACN" metric represent?
A) The maximum number of passengers an aircraft can carry B) The relative impact an aircraft exerts upon the underlying pavement structure
C) The total weight of an aircraft including cargo D) The required length for an aircraft to take off
4. How is the actual runway length calculated after applying corrections for temperature, elevation, and slope?
A) By increasing the basic length by 7% for every 300 m elevation, 1% for every 1°C above standard, and 20% for every 1% of the effective gradient B) By decreasing the basic length by 5% for every 100 m elevation
C) By adding a fixed 10% to the basic length for safety D) By subtracting the take-off distance from the landing distance
5. Which of the following is a primary function of a "time-space diagram" in air traffic control?
A) Calculating the average delay for arriving aircraft B) Visualizing and analyzing aircraft movements on the runway
C) Determining the most efficient route for ground vehicles D) Estimating the required number of baggage carousels
6. What is the purpose of a "fillet" in taxiway design?
A) To increase the speed of aircraft on the taxiway B) To provide an extra wide area at curves to prevent the rear wheel from going off the pavement edge
C) To connect a taxiway to an apron D) To separate parallel taxiways
7. The FAA's four-step approach for forecasting future traffic includes all of the following except:
A) Trip Generation B) Trip Distribution
C) Cost-Benefit Analysis D) Mode-Choice Analysis

8. What is the main purpose of a "travellator" in a large airport?
- A) To transport luggage between terminals B) To provide a convenient way for passengers to cover long distances with ease
- C) To move aircraft from the runway to the apron D) To help ground staff with maintenance tasks
9. In a typical curb lane configuration, what is the purpose of the lane closest to the terminal?
- A) It is a through lane for traffic not accessing the curb B) It is a transition lane for vehicles entering and exiting
- C) It is a quick drop-off/pick-up zone for taxis and private vehicles D) It is a long-term parking area for all vehicles
10. What is the primary function of "Unit Load Devices (ULDs)" in an air cargo facility?
- A) To power ground support equipment B) To provide security screening for cargo
- C) To serve as standardized containers or pallets for transporting cargo D) To store oversized baggage
11. What is the primary purpose of conducting soil classification during pre-construction activities?
- A) To determine the cost of excavation B) To determine the actual bearing capacity of the soil for foundation design
- C) To identify the best location for a batching plant D) To verify the feasibility of the overall site layout
12. Which type of statutory approval is required if an airport is being developed within 500 meters of a coastline?
- A) Change in Land Use (CLU) Approval B) Airport Clearance
- C) Coastal Regulatory Zone (CRZ) Clearance D) Ancient Monument Approval
13. What is the purpose of an impermeable plastic sheet separation membrane in the construction of a rigid pavement?
- A) To act as a drainage layer B) To prevent damage from oil spillage and jet blast
- C) To be provided over the Dry Lean Concrete (DLC) layer before laying the Pavement Quality Concrete (PQC) D) To ensure a smooth riding surface
14. What is the purpose of a Pavement Structural Survey?
- A) To find the surface friction coefficient B) To identify different types of surface defects
- C) To determine the PCN D) To assess the thickness of a pavement's wearing coat
15. A localizer is a VHF radio transmitter and antenna system that operates between what frequencies?
- A) 88.10 MHz and 95.95 MHz B) 108.10 MHz and 111.95 MHz
- C) 118.10 MHz and 125.95 MHz D) 128.10 MHz and 131.95 MHz
16. Which operational measure is used to reduce noise by minimizing the use of idle thrust during descent?
- A) Noise Abatement Departure Procedures B) Continuous Descent Approaches
- C) Single-Engine Taxiing D) Optimized Flight Paths
17. Which type of cargo is transported in pieces separately rather than in a container?
- A) Dry Bulk Cargo B) Liquid Bulk Cargo
- C) Ro-Ro Cargo D) Break Bulk Cargo

18. What is the term for the science of measuring and describing features that affect maritime navigation, such as soundings, shorelines, and tides?
- A) Topographic Survey B) Meteorological Data Collection
C) Hydrographic Survey D) Geological Data Collection
19. Which of the following loads is considered variable or moving in the context of seaports?
- A) Dead Load B) Mooring Load
C) Live Load D) Wind Load
20. What is the difference between Capital Dredging and Maintenance Dredging?
- A) Capital dredging is a continuous process, while maintenance dredging is periodic
B) Capital dredging creates new channels, while maintenance dredging maintains existing depths
C) Capital dredging uses mechanical devices, while maintenance dredging uses hydraulic devices
D) Capital dredging removes naturally disturbed seabed, while maintenance dredging removes undisturbed seabed

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the primary purpose of the three principal domains of a modern airport: the landside, the terminal building, and the airside.	8	L*2	1	1
	b) All arriving aircraft use a common approach path that stretches 10 miles long. Only one type of aircraft utilizes the runway: Type A with a cruising approach speed of 75 miles per hour. Two aircraft are currently lined up for arrival. The minimum safe distance (in miles) to be maintained between trailing and lead flights is 4 miles. Construct the time-space diagram. Assume necessary details.	8	L3	2	1
2.	a) Calculate the corrected final runway length as per ICAO norms based on Elevation, Temperature and Gradient data provided as follows. I) Elevation of the airport site = 185 m II) Standard atmosphere Temperature = 15° C III) Airport reference temp = 30° C IV) Maximum difference in elevation = 10 m V) Basic runway length = 2500 m Assume necessary details	8	L3	1	1
	b) Explain how an airport's runway and taxiway layout can influence its overall airside capacity.	8	L2	2	1
3.	a) Demonstrate an "Airport Master Plan," and what are its main objectives, as outlined by the FAA?	8	L2	1	1
	b) Describe the four-step approach used for forecasting future traffic volumes in airport planning.	8	L2	2	1
Unit – II					
4.	a) Explain the primary purpose of an air cargo apron and list four of its key components.	8	L2	3	1
	b) Illustrate working of the components of the Instrument Landing System to guide an aircraft to a safe landing?	8	L2	4	1
5.	a) Design the terminal curb for the following vehicle demand. The existing curbfront length is 1250 ft. Assume double parking. Design Hour Demand in Vehicles: Private vehicles-300 Taxis-250 Hotel Shuttles-25 Airport Shuttles-20 Other-100 Assume necessary details	8	L3	3	1

b)	Compare the Pavement Structural Survey and the Pavement Surface Survey.	8	L2	4	1
6. a)	Describe the various types of check-in counters found in an airport terminal and how they cater to different passenger categories.	8	L2	3	1
b)	Describe the process for constructing a Dense Bitumen Macadam layer.	8	L2	4	1
Unit – III					
7. a)	Compare the two types of general operations in ports.	8	L2	5	1
b)	Illustrate significance of meteorological data in port design.	8	L2	5	1
8. a)	Describe the function of breakwaters and navigational buoys in a typical port layout.	8	L2	5	1
b)	Explain how a rubble mound breakwater is constructed and the purpose of its different layers.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
